CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A cart is pushed with 20 N and friction is 12 N. The net force is: 1

(a) 12 N **(b)** 8 N **(c)** 32 N **(d)** 20 N

2. A brick of weight 20 N has faces of areas 0.1 square metre and 0.04 square metre. The maximum pressure it can exert when resting on a face is: 1

(a) 200 Pa **(b)** 0.8 Pa **(c)** 500 Pa **(d)** 2 Pa

3. When 10 g of salt is dissolved in 100 mL of water, the volume of the solution is hardly more than 100 mL. This is because: 1

(a) particles of salt fit in the gaps between water particles
(b) salt disappears
(c) water evaporates entirely
(d) salt becomes gas

4. Which of the following pairs contains only compounds? 1

(a) NH_3 and CH_4 **(b)** N_2 and CH_4
(c) O_3 and CO_2 **(d)** Fe and CO_2

5. How many grams of sugar are in 250 g of a 12 per cent sugar solution? 1

(a) 12 g **(b)** 3 g **(c)** 30 g **(d)** 25 g

6. A charged balloon attracts a neutral wall because: 1

(a) the wall is a magnet **(b)** the balloon is heavy
(c) it induces opposite charge near it **(d)** of gravity only

7. A sealed bag of chips puffs up on a hilltop because: 1

(a) air inside cools **(b)** outside pressure is higher
(c) outside pressure is lower than inside **(d)** chips expand

8. A gas can be turned into a liquid if it is cooled below its: 1

(a) boiling point (condensation point) **(b)** melting point only
(c) freezing point only **(d)** room temperature

9. Which of the following is NOT an alloy? 1

(a) bronze **(b)** steel **(c)** brass **(d)** copper

10. Which of the following will show the Tyndall effect? 1

(a) milk **(b)** vinegar **(c)** salt solution **(d)** sugar solution

11. A spring balance shows 3 N for an object. Its mass is about (taking 10 N for 1 kg): 1

(a) 0.3 kg **(b)** 0.03 kg **(c)** 30 kg **(d)** 3 kg

12. A tornado forms mostly: 1

(a) over the deep sea always **(b)** over land under thunderclouds

(c) at the poles **(d)** in caves

13. Rubbing a drop of spirit on the palm gives a cooling feeling because spirit: 1

(a) burns **(b)** is heavy

(c) is cold **(d)** evaporates quickly

14. Electric wires are made of copper because copper is: 1

(a) brittle **(b)** a good conductor and ductile

(c) magnetic **(d)** light and non-conducting

15. In a fair test on the effect of stirring, which must be kept the same? 1

(a) stirring **(b)** time to dissolve

(c) colour **(d)** temperature and amount of sugar

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do athletes use spikes on their shoes? 2

17. How does a hot air balloon use the idea of air expansion? 2

18. State the effect of temperature on the speed of particles and diffusion. 2

19. How many atoms are present in one molecule of C_2H_5OH? 2

20. Why is stirring not changing the solubility of a substance? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the change in arrangement of particles when water vapour changes to ice. 3

22. Give three reasons why gold and silver are used in jewellery. 3

23. Describe the steps for obtaining crystals of copper sulphate from its saturated solution. 3

24. A bar magnet is dipped in iron filings. Where do most filings stick and why? 3

OR

How does a lift pump or dropper make use of air pressure? Explain one. 3

25. Discuss the effect of area of surface on the rate of evaporation with an example. 3

OR

Why are metals used for making bells, utensils and wires? Give properties. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe an experiment to find out how the solubility of salt and of sugar changes with temperature. State the variable, procedure, table and conclusion. 5

OR

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Discuss the effect of air resistance on a falling feather, a stone and a parachutist. 5

27. Describe the safety measures a community should take before, during and after a cyclone. 5

OR

Describe how pressure and temperature can be used to liquefy gases and give three uses. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND